MODULE 4 · RUNWAYS AND PERFORMANCE

Lesson 16: Take-off on Contaminated Runways

Lesson 16 of 30 · about 9 min

LEARNING OBJECTIVES

- Apply AFM contaminant depth and crosswind limits for take-off.
- Explain aquaplaning and estimate aquaplaning speed.
- Plan for a rejected take-off and directional control on a slippery runway.



Take-off on a contaminated runway needs contaminated-runway performance data — and a clear plan for a reject.

Before you start

- Obtain the latest RCR. Confirm that the contaminant type and depth are within the AFM limits for take-off.
- Calculate performance using the contaminated runway data and the reported RWYCC: no assumed-temperature reduced thrust, and possibly a lower V1 and mass.
- Apply the reduced crosswind limit for the runway condition (Operations Manual / FCOM table).
- Brief the rejected take-off: decision speed, directional control technique, and the effect of reverse thrust in a crosswind.

Aquaplaning

When a tyre rides on a layer of water or slush, it loses contact with the runway. Braking and steering become almost useless. The approximate minimum speed at which **dynamic aquaplaning** starts is:

RULE OF THUMB

V_P (kt) ≈ 9 × √(tyre pressure in psi) for a rotating tyre (take-off). For a tyre that is not yet rotating, as at touchdown, the speed is lower — about 7.7 × √(tyre pressure).

Example: tyre pressure 196 psi → √196 = 14 → about 126 kt for a rotating tyre, about 108 kt at touchdown.

Other forms include **viscous aquaplaning** (a thin film on a smooth or rubber-coated surface) and **reverted rubber aquaplaning** (a locked wheel overheating and producing steam).

During the take-off roll

- Where the AFM requires it, carry out a static run-up to shed fan ice. On a slippery surface the brakes may not hold, so use the technique in the FCOM.
- Apply thrust symmetrically. Keep straight with the rudder, using small nosewheel steering inputs at low speed only.
- Expect slower acceleration in slush or wet snow, and spray striking the airframe.

Rejected take-off on a slippery runway

- Stopping distances are much longer. Up to V1, reject for the reasons in your SOPs. Above V1, continue.
- Use maximum braking (or the RTO autobrake) and maximum reverse thrust as the procedure states.
- **Reverse thrust in a crosswind** pushes the aircraft downwind and makes it weathervane. If the aircraft starts to drift off the centreline, the typical technique is: reduce reverse thrust towards idle, release the brakes momentarily, regain the centreline with rudder and steering, then reapply braking and reverse. Follow your FCOM.

After take-off

Slush and snow thrown onto the landing gear, flaps and wheel wells can freeze at altitude. Some AFMs recommend specific actions after a take-off from a slush-covered runway, such as delayed retraction or cycling the gear. Apply them only when your FCOM states them.

KEY TAKEAWAYS

- Check contaminant type and depth against AFM limits; use contaminated runway performance.
- Aquaplaning speed ≈ 9 × √(tyre pressure) for a rotating tyre, lower at touchdown.
- In a crosswind RTO, reverse thrust can push you downwind — know the recovery technique.

KNOWLEDGE CHECK

Tyre pressure is 144 psi. Approximately at what speed could dynamic aquaplaning begin during the take-off roll?

A. About 72 kt

B. About 108 kt

C. About 144 kt

D. About 160 kt

Answer: B. 9 × √144 = 9 × 12 = 108 kt for a rotating tyre.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.